

```
1  #include <stdio.h>
2
3
4  #define MAX_SIZE 10
5
6  typedef struct {
7      int data[MAX_SIZE];
8      int size;
9  } arrayADT;
10
11 void display (arrayADT *arr);
12
13 void init (arrayADT *arr) {
14     int size;
15
16     // int num;
17     //
18     // printf("Enter number of elements: ");
19     // scanf("%d", &size);
20     //
21     // for(int i = 0; i < size; i++){
22     //     printf("Enter element %d: ", i + 1);
23     //     scanf("%d", &num);
24     //     printf("\n");
25     //     arr -> data[i] = num;
26     // }
27
28     arr -> size = 0;
29
30 }
31
32 void delete (arrayADT *arr, int index) {
33     if (index < 0 || index >= arr -> size || arr -> size >=
        MAX_SIZE) {
34         printf("Invalid index or Element does not exist\n");
35         return;
36     }
```

```
37
38     for (int i = index; i < arr -> size - 1; i++){
39         arr -> data[i] = arr -> data[i + 1];
40     }
41     arr -> size--;
42     display(arr);
43 }
44
45 void insertAt (arrayADT *arr, int index, int value) {
46     if (index < 0 || index > arr -> size || arr -> size >=
47         MAX_SIZE) {
48         printf("invalid index or list is full\n");
49         return;
50     }
51     for (int i = arr -> size; i > index; i--) {
52         arr -> data[i] = arr -> data[i-1];
53     }
54     arr -> data[index] = value;
55     arr -> size++;
56     display(arr);
57 }
58
59 void insertEnd (arrayADT *arr, int value) {
60     if (arr -> size >= MAX_SIZE) {
61         printf("The List is full\n");
62         return;
63     }
64
65     arr -> data[arr -> size] = value;
66     arr -> size++;
67     display(arr);
68 }
69
70 void display (arrayADT *arr) {
71     if (arr -> size == 0) {
72         printf("List is Empty\n");
```

```
73         return;
74     }
75
76     for (int i = 0; i < arr -> size; i++) {
77         printf("%d ", arr -> data[i]);
78     }
79     printf("\n");
80 }
81
82
83 int main() {
84
85     arrayADT arr;
86     init(&arr);
87     insertAt(&arr, 0, 59);
88     insertEnd(&arr, 7);
89     insertAt(&arr, 1, 16);
90     insertEnd(&arr, 8);
91     delete(&arr, 3);
92
93     return 0;
94 }
95
```